

Pre-read and Post-read materials

Lecture: Advanced Prompt Engineering

Pre-read

Read these before the lecture. They are short and practical, covering the basics you will need so the session can move quickly into advanced techniques.

Getting started with prompting:

- Prompt Engineering Guide: <https://www.promptingguide.ai/>
- Learn Prompting: <https://learnprompting.org/docs/advanced/introduction>
- Anthropic Prompt Engineering Overview: <https://docs.anthropic.com/en/docs/build-with-claude/prompt-engineering/overview>
- OpenAI Prompt Engineering Guide (six simple tactics for better prompts): <https://platform.openai.com/docs/guides/prompt-engineering>

Decoding parameters (temperature, top-p, top-k):

Several techniques in this lecture (especially Self-Consistency) depend on understanding why a model's output changes from run to run. These readings explain the settings that control that.

- Temperature, Top-p and Top-k explained (a clear, plain-English walkthrough of each setting): https://www.f22labs.com/blogs/what-are-temperature-top_p-and-top_k-in-ai/
- Understanding LLM Sampling: Top-K, Top-P, and Temperature (short conceptual overview): <https://towardsai.net/p/machine-learning/understanding-llm-sampling-top-k-top-p-and-temperature>
- OpenAI API reference for the parameters (temperature, top_p, and related settings): <https://platform.openai.com/docs/api-reference/chat/create>

Post-read

(for those who wish to explore further after the lecture)

For a deeper, research-level understanding, refer to the foundational papers below. The easiest way is to search the titles on <https://scholar.google.com>.

- Brown et al. (2020). Language Models are Few-Shot Learners (the GPT-3 paper that established in-context learning and the few-shot prompting paradigm). **NeurIPS**. arXiv:2005.14165.
- Wei et al. (2022). Chain-of-Thought Prompting Elicits Reasoning in Large Language Models (the original CoT paper; introduces few-shot CoT and shows it emerges with scale). **NeurIPS**. arXiv:2201.11903.

- Kojima et al. (2022). Large Language Models are Zero-Shot Reasoners (the "Let's think step by step" paper; zero-shot CoT). **NeurIPS**. arXiv:2205.11916.
- Wang et al. (2023). Self-Consistency Improves Chain of Thought Reasoning in Language Models (sample diverse paths, majority vote). **ICLR**. arXiv:2203.11171.
- Zhang et al. (2023). Automatic Chain of Thought Prompting in Large Language Models (Auto-CoT; clustering plus Zero-Shot-CoT for exemplar generation). **ICLR**. arXiv:2210.03493.
- Yao, Yu, Zhao, Shafran, Griffiths, Cao, Narasimhan (2023). Tree of Thoughts: Deliberate Problem Solving with Large Language Models (ToT; propose, evaluate, and search over thought nodes; Game of 24 benchmark). **NeurIPS**. arXiv:2305.10601.
- Yao, Zhao, Yu, Du, Shafran, Narasimhan, Cao (2023). ReAct: Synergizing Reasoning and Acting in Language Models (interleaved Thought, Action, Observation trajectories on HotpotQA and AlfWorld). **ICLR**. arXiv:2210.03629.
- Zhang, Yuan, Yao (2023). Meta Prompting for AI Systems (structure-oriented prompting framework that asks the model to reason about its own prompting strategy). arXiv:2311.11482.
- Li, Peng, He, Galley, Gao (2023). Guiding Large Language Models via Directional Stimulus Prompting (a tuneable policy LM generates keyword "hints" to steer a frozen black-box LLM; trained with RL using ROUGE as reward). **NeurIPS**. arXiv:2302.11520.
- Greshake, Abdelnabi, Mishra, Endres, Holz, Fritz (2023). Not what you've signed up for: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection (the foundational paper on indirect prompt injection; the Bing Chat example from the slides comes from this work). **AISeC**. arXiv:2302.12173.
- Schulhoff et al. (2024). The Prompt Report: A Systematic Survey of Prompting Techniques (the most comprehensive prompting survey to date; a taxonomy of 58 text-only techniques and 40 for other modalities). arXiv:2406.06608.
- Huang et al. (2023). A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions (the standard reference for the hallucination taxonomy). **ACM TOIS**. arXiv:2311.05232.